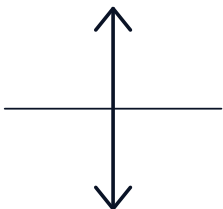
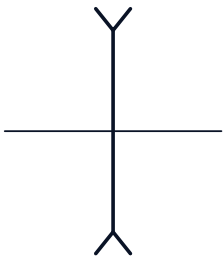
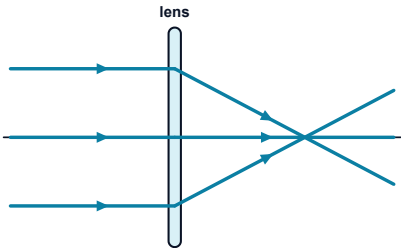
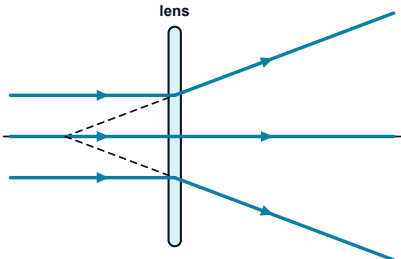


Lenses and visible light

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Name the process that happens to light in a lens.	Refraction	☐
2 ★	Describe what a convex lens does to parallel rays of light.	It brings them together at the principal focus.	☐
3	What is the focal length of a lens?	The distance from the lens to the principal focus.	☐
4	Describe what a concave lens does to parallel rays of light.	It spreads them out, as if from the principal focus.	☐
5 ★	What is a real image?	An image formed where rays of light actually meet.	☐
6 ★	What is a virtual image?	An image that rays of light only appear to come from.	☐
7	Name the type of image that can be shown on a screen.	A real image	☐
8	Name the type of lens that produces only virtual images.	A concave lens	☐
9	Name the type of lens that this symbol represents. 	A convex lens	☐

	QUESTION	ANSWER	✓
10	Name the type of lens that this symbol represents. 	A concave lens	☐
11 ★	Name the type of lens shown in this ray diagram. 	A convex lens	☐
12	Name the type of lens shown in this ray diagram. 	A concave lens	☐
13	State how different colours of light differ.	They have different wavelengths.	☐
14 ★	What is specular reflection?	Reflection from a smooth surface in a single direction.	☐
15	What is diffuse reflection?	Reflection from a rough surface, scattering light in many directions.	☐
16	Explain how colour filters work.	They transmit some wavelengths and absorb the rest.	☐
17	Explain why blue objects appear blue.	They reflect blue light and absorb the other colours.	☐
18	Explain why white objects appear white.	They reflect all colours of light equally.	☐

QUESTION		ANSWER	✓
19	Explain why black objects appear black.	They absorb all colours of light.	☐
20	In words, state the equation for magnification.	magnification = image height ÷ object height	☐
21	State the unit of magnification.	It has no unit (it is a ratio).	☐
22	State the types of image that a convex lens can produce.	Real or virtual	☐
23	State what determines the colour of an opaque object.	The wavelengths of light it reflects most strongly.	☐
24	State what an opaque object does to light.	It absorbs or reflects it; none is transmitted.	☐
25 ★	A blue object is viewed in red light. State and explain the colour it appears.	Black. It absorbs the red light and has no blue light to reflect.	☐